

KRIS LAI

Senior AI Engineer

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 kriskirla

EXPERIENCE

Senior AI Engineer | SleekFlow Technologies Limited Aug 2025 – Present

- Architected a production-grade ReAct engine enabling non-technical users to deploy rule-bound agents via a multi-stage Plan-Draft-Self-Review pipeline. Balanced deterministic state machines with non-deterministic LLM reasoning, enforcing zero-fail structured schemas with recursive self-healing to drastically minimize hallucinations.
- Designed empirically validated mechanisms to mitigate agent biases, including Lost in the Middle effects and LLM laziness. Engineered multi-agent consensus protocols to resolve execution deadlocks alongside robust episodic/semantic memory systems featuring asynchronous state hydration for long-horizon, human-in-the-loop workflows.
- Built high-precision pipelines for multi-modal ingestion (docs, web data, live search) using cross-modal alignment, delta syncs, and hybrid search fusion. Solved large-scale tool selection bottlenecks through two-tier tool retrieval, multi-hop reasoning, and type-safe argument synthesis.
- Engineered internal AI development workflows utilizing context-window optimization and sub-agent orchestration, accelerating team velocity by over 150%. Established continuous LLM-as-a-judge trajectory auditing while optimizing token budgets and latency through dynamic model cascading and prompt caching.

Senior Software Engineer | Morningstar, Inc Jan 2022 – July 2025

- Architected and led the migration of a core portfolio analytics engine to a high-scale, cross-region AWS serverless architecture (ECS, Fargate, Lambda), enabling seamless support for large-scale portfolio analysis on the Morningstar Workstation Platform.
- Designed high-performance .NET analytics services that utilize DynamoDB and event-driven patterns, drastically improving the performance of complex portfolio risk scoring while maintaining cost efficiency.
- Established operational excellence by implementing enterprise-grade disaster recovery (Route 53, Global DynamoDB), IaC via Terraform, and automated security/compliance gates (SOC 2, SonarQube, Checkmarx) to ensure system reliability and code quality.

Full Stack Software Engineer | Peak Power Inc Jul 2020 – Jan 2022

- Engineered a scalable full-stack analytics platform (Angular, Spring Boot, Java) enabling enterprise customers to derive actionable insights from real-time energy, occupancy, and climate telemetry.
- Architected a high-throughput, multi-threaded Java driver for global IoT connectivity that improved system latency by 5x, leveraging Kafka, Python, and Docker to build robust stream processing, automated alerting, and full-stack observability (Datadog, Sentry).

Software Development Team Lead | IBM Feb 2019 – Jul 2020

- Pioneered "Project ARBIT," architecting a standalone Python/Go RESTful microservice containerized with Docker to automate Db2 failure investigations, reducing annual operational overhead by approximately \$1M.
- Led a cross-functional functional verification team of 11 developers and QA engineers, streamlining release pipelines to transition core Db2 components from development to production with minimal error rates.

CERTIFICATIONS

- AWS Certified Solutions Architect - Associate | AWS  Nov 2022 – Nov 2025
- Enterprise Design Thinking - Team Essentials for AI | IBM  Jun 2019
- Enterprise Design Thinking Co-Creator | IBM  Jun 2019

EDUCATION

Bachelor of Science (B.Sc.) | University of Toronto 2013 – 2018

- Computer Science - Specialist in Software Engineering

LANGUAGES

English (Native), Cantonese (Native),
Mandarin (Fluent)